

Mahi Pono & East Maui Water

Cori Mackie

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Individual Research Paper

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1. Introduction and Position

In 1869, Samuel Alexander and Henry Baldwin purchased arid land in Central Maui and helped develop the 17-mile Hāmākua Ditch to divert water from East Maui streams. The system eventually expanded into a network of intakes, ditches, tunnels, and reservoirs that transformed dry Central Maui land into productive agricultural acreage (“Battle Looms,” 1997). These diversions also reduced water available to East Maui communities, where nearly 500 lo‘i were claimed in Ke‘anae and Wailuanui at the time of the Māhele (Group 70 International et al., 1994, as cited in “Battle Looms,” 1997).

In 1876, the government granted water access for \$100 per year for 20 years (“Battle Looms,” 1997). This raises an economic question that remains relevant today: **Does the price paid for access to East Maui water reflect the economic value that water creates?**

The question is particularly relevant now, and having grown up on Maui, I have a vested interest in its future. Record-low stream flows in 2025 (Hawai‘i Department of Land and Natural Resources [DLNR], 2025b), a July 2026 court ruling limiting diversions (Kubota, 2026), and the Board of Land and Natural Resources’ (BLNR’s) August 2026 rejection of a proposed 30-year water license (Perry, 2026) have increased uncertainty over how Maui’s limited water should be allocated.

My hypothesis is that the price paid for East Maui water does not fully reflect its economic value. I will examine what Mahi Pono pays for access, the agricultural value that water supports, and what Maui gives up when that water is diverted, using **scarcity, supply and demand, opportunity cost, value capture, and market structure**.

2. Background: The Permit and Who Holds It

BLNR sets the price for East Maui water through annual revocable permits. The 2026 permit was issued to Mahi Pono and East Maui Irrigation (EMI) after Mahi Pono acquired full ownership of EMI in June 2025 (DLNR, 2025a). EMI supplies Mahi Pono's Central Maui farms, Maui County's Upcountry system, and other agricultural and municipal users. Mahi Pono plants around 12,900 acres of the approximately 35,000 in Central Maui in 2026, and some is used for cattle (Mahi Pono, 2026). Mahi Pono owns the EMI infrastructure, but the water remains a public resource regulated by the State. In July 2026, the court set an interim diversion cap of 31.5 million gallons per day (Kubota, 2026), and in August BLNR rejected a proposal to auction a 30-year water license (Perry, 2026).

3. Price vs. Value

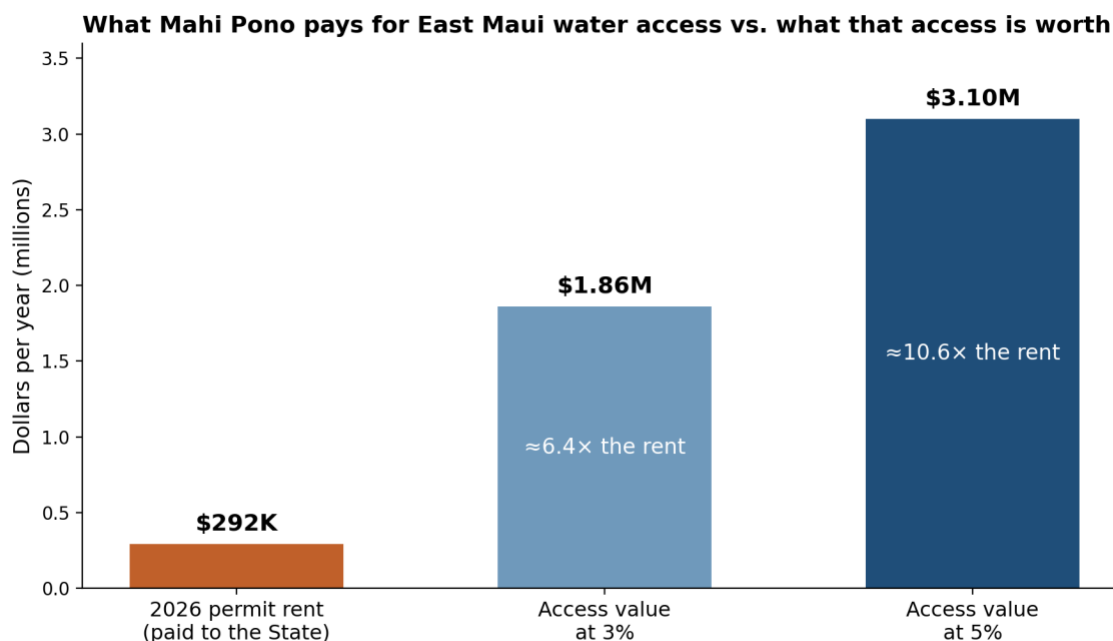
I use 3% as a conservative benchmark based on the approximately 3% 10-year U.S. Treasury yield in 2018 (Board of Governors of the Federal Reserve System, n.d.). I also test 5% to show how the estimated annual value changes under a higher assumed rate of return.

If the \$62 million holdback represents the market value of secure water access, that access is worth about \$1.9–\$3.1 million annually (Alexander & Baldwin, 2019), compared with about \$292,000 in annual permit rent (DLNR, 2025a). The gap—about \$1.6–\$2.8 million, or 6–11 times the rent—is economic rent: value created by access to a public resource priced below its estimated value. A&B captured part of that value in the sale, while Mahi Pono benefits through lower water costs and productive land. The public receives about \$292,000 annually.

The \$62 million reflects secure, long-term access. A one-year permit may not fully reflect the economic value of that reliable access.

Figure 1

Annual Permit Rent vs. Annual Value of Water Access, 2026



Access value = the ~\$62 million of the 2018 sale price held back pending water access, converted to a yearly value at an assumed 3% or 5% rate. Sources: Alexander & Baldwin (2019), Form 10-K for 2018; Hawai'i DLNR (2025), Board submittal D-15 (\$24,306/month).

Note. Access value is the approximately \$62 million held back in the 2018 sale pending water access, converted to a yearly value at an assumed 3% or 5% rate. Data from Alexander & Baldwin (2019) and DLNR (2025a).

4. Who Pays What

The price of water varies by user. Mahi Pono pays approximately 2.5 cents per 1,000 gallons in permit rent (DLNR, 2025a). Including EMI operating and maintenance costs raises its estimated all-in cost to about 26 cents per 1,000 gallons (Greenaway, 2026). By comparison, Maui County's agricultural water rate is \$1.31 per 1,000 gallons (County of Maui Department of Water Supply, 2026).

5. What Maui Gives Up

Historical stream-flow data show that Maui's water supply is not constant. Statewide median base flow declined about 23% after 1943 (Bassiouni & Oki, 2013), while Honopou's

average flow declined about 14% over the same period (Figure 2). Its 2025 average was the second lowest since records began in 1912 (U.S. Geological Survey, n.d.).

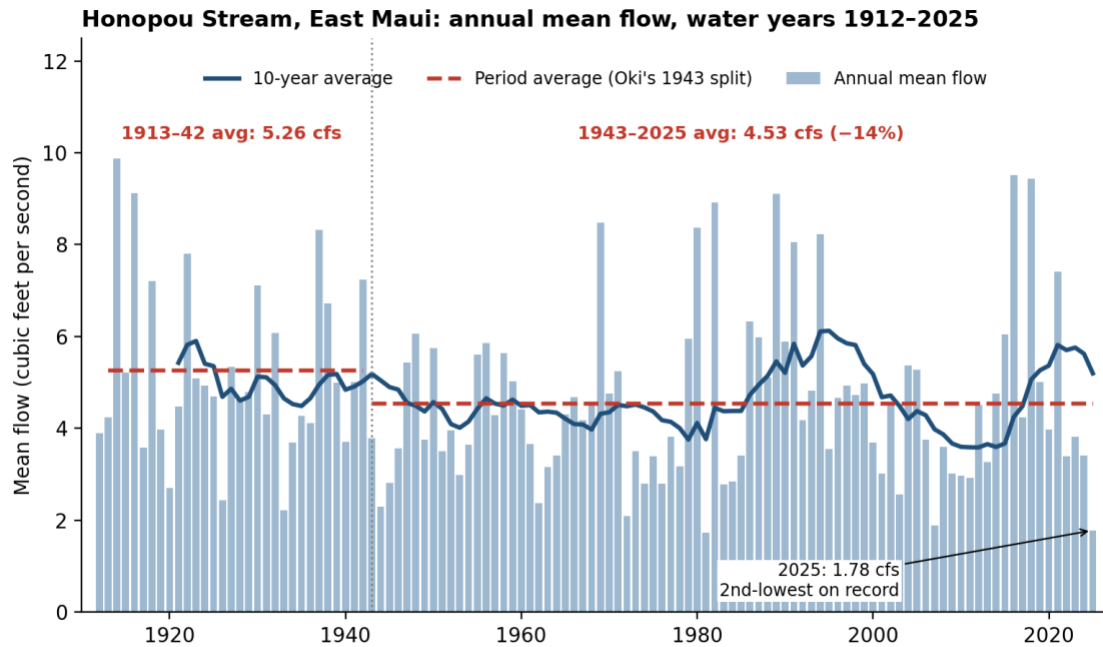
This creates scarcity when water supply declines, but the diversion cap remains 31.5 million gallons per day. If supply fell another 23%, the permit would represent a larger share of the remaining resource. **The permit does not become larger, but its claim on a shrinking resource does.**

As water becomes scarcer, its economic value increases, but the permit price does not automatically adjust. The opportunity cost falls on competing uses, including agriculture, communities, and stream ecosystems.

The 23% scenario is not a prediction. Stream flow varies with rainfall, drought, climate, groundwater recharge, and diversion practices; it simply illustrates the effect of a fixed diversion on a shrinking resource.

Figure 2

Annual Mean Flow of Honopou Stream, East Maui, Water Years 1912–2025



Source: USGS gauge 16587000, annual statistics (retrieved Oct. 5, 2026). The 1943 split follows Bassiouni & Oki (2013).

Note. Data from U.S. Geological Survey (n.d.), gauge 16587000. The 1943 split follows Bassiouni and Oki (2013).

6. Mahi Pono's Best Case

Mahi Pono took over former sugar lands with an uncertain agricultural future and has invested in new crops, agricultural jobs, and local food production. It also operates and maintains the aging EMI system, which serves agriculture and Maui County users.

7. Recommendation

In the short term, I recommend that the State replace the fixed permit rent with a pricing model that responds to water scarcity. When stream flows decline, the price of diverted water should increase to reflect the value of a shrinking resource.

This model should recognize Mahi Pono's contributions, including agricultural jobs, local food production, and the approximately \$3 million annual cost of operating EMI. The system

also serves Maui County. Even including these costs, Mahi Pono's estimated all-in water cost is still well below the County's farm rate.

Longer term, the State should reduce dependence on a privately controlled irrigation system by separating water allocation from EMI's ownership of the infrastructure. This would mean creating a new future from a structure more than 150 years old that was set up to profit a few and doesn't mean a sentence for eternity. This would address the natural-monopoly problem created when one private entity controls the primary system used to transport a public resource. A public buyer exists; Maui voters created the East Maui Water Authority in 2022 with 64% of the vote (Scheuer, 2026). In 1876, the attorney general recommended that the government could buy the ditch back "upon payment of the actual cost thereof" ("Battle Looms," 1997, "Grass Grows, Water Runs" section).

A scarcity-based price, combined with greater public control over future water allocation, could preserve agricultural access while better reflecting the value and competing uses of East Maui water.

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